

GDPNow Methodology: A Reproducible, Math-Forward Guide

A tracking-model nowcast of U.S. real GDP growth

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Abstract

GDPNow is the Federal Reserve Bank of Atlanta’s “tracking-model” nowcast of real GDP growth. It mimics core features of the U.S. Bureau of Economic Analysis (BEA) production process by forecasting key GDP expenditure subcomponents using monthly source data and then aggregating the component forecasts using Fisher chain-weighting. This document provides a math-forward methodology summary based on Higgins (2014) and subsequent Atlanta Fed “modifications” memos, and it outlines the data and computation needed to replicate a GDPNow-style nowcast in Python.

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1 High-level overview

GDPNow is best viewed as a running estimate of quarterly real GDP growth (reported at a seasonally adjusted annual rate, SAAR) updated as new high-frequency economic releases arrive. It is a mechanical model: published results are derived purely from mathematical calculations with no judgmental adjustments.

The baseline framework in Higgins (2014) is organized as six steps:

1. **Quarterly BVAR backbone:** forecast 13 high-level GDP quantity components with a quarterly Bayesian VAR (BVAR), using data only through the most recent official GDP quarter.
2. **Monthly activity factor:** estimate a dynamic factor from a large panel of monthly indicators (jagged edge, missing observations).
3. **Monthly forecasting:** forecast missing monthly source data with factor-augmented autoregressions (FAARs), then aggregate completed monthly paths to quarterly growth rates.
4. **Bridge equations:** forecast granular investment and government subcomponents using monthly indicators; aggregate to coarser components and combine with the quarterly BVAR.
5. **Special modules:** handle PCE, net exports, and inventories using bespoke routines.
6. **Chain-weighting:** combine component forecasts into a GDP nowcast using Fisher chain-weighting (as in the BEA methodology).

2 Notation and timing

Let t index quarters and let $m \in \{1, 2, 3\}$ index the month within quarter t . For a monthly series $x_{t,m}$, define the quarterly average level

$$X_t \equiv \frac{1}{3} (x_{t,1} + x_{t,2} + x_{t,3}). \quad (1)$$

We use $\Delta \log(\cdot)$ for log differences. If g_t denotes quarterly (non-annualized) growth, the SAAR growth rate is

$$\text{SAAR}_t = 100 \left((1 + g_t)^4 - 1 \right). \quad (2)$$

GDPNow is run “as of” a particular date with a jagged-edge monthly dataset: some month- t values are observed, some are missing because their releases have not occurred yet, and some are revised.

3 Step 1: Quarterly BVAR for GDP subcomponents

GDPNow forecasts 13 real (quantity) GDP components with a quarterly BVAR with 5 lags. Let $\mathbf{Q}_t \in \mathbb{R}^{13}$ collect log quantities of the components. The BVAR can be written schematically as

$$\mathbf{Q}_t = \mathbf{c} + \sum_{\ell=1}^5 A_\ell \mathbf{Q}_{t-\ell} + \boldsymbol{\eta}_t, \quad (3)$$

estimated with shrinkage priors (Minnesota-style dummy-observation implementation). The one-step-ahead forecast for component i is denoted

$$\Delta \widehat{\log Q_{t+1,i}}^{\text{BVAR}}. \quad (4)$$

GDPNow also runs a companion quarterly BVAR for the 13 implicit price deflators.

Role of the BVAR. The quarterly BVAR provides a stable low-dimensional forecast anchor. For several components it is combined with bridge-equation forecasts built from monthly data.

4 Step 2: Prices and the monthly dynamic factor

4.1 Step 2a: Monthly deflators and imputation

Many monthly source series are nominal and must be deflated. In some cases nominal series are released before the needed deflator; GDPNow forecasts missing deflators using a multi-variable monthly price BVAR and conditional forecasting methods to account for staggered releases.

2025 modification: imputing missing monthly price observations. When a monthly price observation is missing but adjacent months are available (e.g., due to a government shutdown), the Atlanta Fed documentation describes imputing the missing observation by a geometric mean. If p_{Sep} and p_{Nov} are available but p_{Oct} is missing, set

$$p_{\text{Oct}} \leftarrow \sqrt{p_{\text{Sep}} p_{\text{Nov}}}. \quad (5)$$

4.2 Step 2b: Dynamic factor model

Let $z_{t,m,i}$ be a standardized stationary transformation of monthly series i from a large panel ($N \approx 124$ in the original paper, later 126). GDPNow estimates a one-factor dynamic factor model:

$$f_{t,m} = \rho_1 f_{t,m-1} + \rho_2 f_{t,m-2} + \rho_3 f_{t,m-3} + u_{t,m}, \quad (6)$$

$$z_{t,m,i} = \gamma_i f_{t,m} + \varepsilon_{t,m,i}. \quad (7)$$

The factor $f_{t,m}$ is extracted in a state-space framework with Kalman filtering/smoothing. Missing observations from jagged-edge releases are handled naturally in the Kalman filter.

2018 modification: serial correlation in idiosyncratic errors. To mitigate predictable jumps around certain releases (notably ISM manufacturing), the Atlanta Fed documentation allows idiosyncratic components for some series to follow AR(1) dynamics:

$$\varepsilon_{t,m,i} = \phi_i \varepsilon_{t,m-1,i} + v_{t,m,i}. \quad (8)$$

2020 modification: no outlier replacement in factor inputs. Prior to April 2020, some extreme observations in the factor panel were replaced using an interquartile-range rule. Beginning April 2020, the panel is used as-is (including large pandemic-era movements).

5 Step 3: Forecast missing monthly source data (FAARs)

For many monthly series $y_{t,m}^{(i)}$ used as GDP “source data” (after any needed deflation), GDPNow forecasts missing months with a factor-augmented autoregression:

$$\Delta \log y_{t,m}^{(i)} = \alpha_i + \sum_{k=1}^{q_i} \gamma_{i,k} \Delta \log y_{t,m-k}^{(i)} + \sum_{j=0}^{r_i} \beta_{i,j} f_{t,m-j}. \quad (9)$$

Lag orders (q_i, r_i) are chosen by an information criterion over bounded sets.

2022 modification: COVID-era dummies and shorter lag bounds. Beginning in 2022, the monthly forecasting equation includes a set of dummy variables for the months immediately following March 2020 to better fit pandemic-era breaks, and the maximum lag length searched by the information criterion is reduced.

5.1 Monthly-to-quarterly aggregation

Given observed and forecasted monthly values within quarter t , define the completed quarterly average

$$\tilde{Y}_t^{(i)} = \frac{1}{3} \sum_{m=1}^3 \tilde{y}_{t,m}^{(i)}, \quad (10)$$

and the implied quarterly growth

$$\widehat{\Delta \log Y_t^{(i)}} = \log \tilde{Y}_t^{(i)} - \log \tilde{Y}_{t-1}^{(i)}. \quad (11)$$

These quarterly-growth objects feed the bridge equations.

6 Step 4: Bridge equations (investment and government)

6.1 Granular bridge equations

For several GDP components (notably fixed investment and government spending), GDPNow forecasts “granular” subcomponents using bridge equations and then aggregates them. Let $X_t^{(g)}$ denote a granular quarterly subcomponent and let $\widehat{\Delta \log Y_t^{(\ell)}}$ denote the quarterly growth of related monthly indicator(s) from Step 3. The first-stage bridge equation is

$$\Delta \log X_t^{(g)} = \varphi_g + \sum_{\ell=1}^{n_g} \theta_{g,\ell} \widehat{\Delta \log Y_t^{(\ell)}}. \quad (12)$$

If no suitable monthly indicator exists, the model uses a univariate AR forecast for the subcomponent.

6.2 Aggregation across granular subcomponents

Let $w_{t-1}^{(g)}$ be a nominal expenditure share of granular subcomponent g in quarter $t - 1$. GDPNow forms an approximation to Fisher aggregation by using a Tornqvist-weighted growth-rate average:

$$\widehat{\Delta \log X_t}^{\text{bridge}} = \sum_{g \in \mathcal{G}} w_{t-1}^{(g)} \widehat{\Delta \log X_t^{(g)}}. \quad (13)$$

2025 modification: office-structures split and data-center deflator. Beginning with 2020 data (after a BEA revision) that disaggregates private nonresidential office structures into “data centers” and “general, financial, and other” structures, the Atlanta Fed documentation uses a separate price deflator for data centers constructed as a Tornqvist-weighted geometric mean of industrial-building and warehouse-building PPIs:

$$\log P_{t,m}^{\text{DC}} = s_{t,m}^{\text{ind}} \log P_{t,m}^{\text{ind}} + (1 - s_{t,m}^{\text{ind}}) \log P_{t,m}^{\text{wh}}, \quad (14)$$

where $s_{t,m}^{\text{ind}}$ is based on nominal Tornqvist investment shares.

6.3 Combining bridge and BVAR forecasts

For several components (residential investment, equipment, IPP, nonresidential structures, federal government spending, and state and local government spending), GDPNow combines the quarterly BVAR forecast and the bridge forecast using a restricted regression (no intercept):

$$\Delta \log X_t = \delta \widehat{\Delta \log X_t}^{\text{BVAR}} + (1 - \delta) \widehat{\Delta \log X_t}^{\text{bridge}} + e_t, \quad 0 \leq \delta \leq 1. \quad (15)$$

2022-2023 modifications: exclude pandemic quarters; recency-weighted restricted LS.

To reduce distortion from extreme pandemic observations, the documentation describes dropping the 2020Q1–2020Q4 observations when estimating some combination regressions. In addition, beginning 2023, estimation uses recency-weighted least squares and enforces the “weights sum to one” restriction explicitly. Let X be the $T \times 2$ regressor matrix with the BVAR and bridge forecasts and Y the $T \times 1$ vector of outcomes. Let W be diagonal with exponentially decaying weights (higher weight on recent quarters), e.g.

$$w_h \propto \left(\frac{\lambda}{1 + \lambda} \right)^h, \quad h = 0, 1, 2, \dots, \quad (16)$$

where h indexes how many quarters an observation is before the most recent official GDP quarter. The unrestricted WLS estimator is

$$\hat{\delta}^{\text{WLS}} = (X'WX)^{-1}X'WY, \quad (17)$$

and the restricted WLS estimator under $R\delta = q$ with $R = [1 \ 1]$ and $q = 1$ is

$$\hat{\delta}^{\text{RWLS}} = \hat{\delta}^{\text{WLS}} - (X'WX)^{-1}R'(R(X'WX)^{-1}R')^{-1}(R\hat{\delta}^{\text{WLS}} - q). \quad (18)$$

If the resulting weight on either forecast exceeds 1.0, the procedure caps that weight at 1.0 and sets the other weight to 0.

7 Step 5: Net exports (goods and services)

GDPNow uses a specialized approach for net exports. Goods imports and exports are deflated into real measures using import and export price indices. The model forecasts the contribution of net exports to GDP growth and reconciles that forecast with separate forecasts of imports and exports.

7.1 Contribution-style forecasting and reconciliation

An approximate monthly contribution for month m in quarter t is

$$\text{cont}_{t,m}^{\text{NX}} \approx s_{t,m-1}^{\text{exp}} \Delta \log \text{rgexp}_{t,m} - s_{t,m-1}^{\text{imp}} \Delta \log \text{rgimp}_{t,m}, \quad (19)$$

where shares are based on lagged nominal levels relative to nominal GDP. GDPNow forecasts $\text{cont}_{t,m}^{\text{NX}}$ with a FAAR and then applies an adjustment to ensure that the separate export and import forecasts imply the targeted contribution.

2025 modification: subtract nonmonetary gold; special handling after AEI. Beginning in 2025, the documentation describes modifying goods trade inputs by subtracting nonmonetary gold trade from BOP-based goods trade measures. If $M_{t,m}^{\text{BOP}}$ and $X_{t,m}^{\text{BOP}}$ are BOP imports and exports and $M_{t,m}^{\text{gold}}, X_{t,m}^{\text{gold}}$ are the corresponding nonmonetary gold series, then

$$M_{t,m}^{\text{ex gold}} = M_{t,m}^{\text{BOP}} - M_{t,m}^{\text{gold}}, \quad X_{t,m}^{\text{ex gold}} = X_{t,m}^{\text{BOP}} - X_{t,m}^{\text{gold}}. \quad (20)$$

When the AEI report provides month- t Census-basis goods trade aggregates but does not disentangle gold in the “industrial supplies and materials” category, the documentation describes discarding that subaggregate and using a small Bayesian VAR (with conditional forecasting) to infer the month- t value of industrial supplies and materials trade excluding gold.

8 Step 6: Personal consumption expenditures (PCE)

PCE is handled differently because much of PCE is already monthly. GDPNow partitions PCE into subcomponents (goods and services), forecasts missing months with FAARs, and uses additional high-frequency signals for certain categories (e.g., retail control, motor vehicles).

9 Step 7: Change in private inventories (CIPi)

Inventory investment is one of the most complex pieces. At a high level, GDPNow approximates monthly inventory investment in industry i as

$$\text{CIPi}_{t,m}^{(i)} = \Delta \text{Invent}_{t,m,\text{book}}^{(i)} + \text{IVA}_{t,m}^{(i)}, \quad (21)$$

where $\Delta \text{Invent}_{t,m,\text{book}}^{(i)}$ is the published book-value change in inventories and $\text{IVA}_{t,m}^{(i)}$ is an inventory valuation adjustment (IVA). Because IVAs are not observed in real time for the current quarter, the model forecasts them using producer prices and BVAR-based auxiliary systems. Recent documentation also extends pandemic-era estimation adjustments and recency weighting to some inventory-related combination regressions.

10 Step 8: Fisher chain-weighting aggregation to real GDP growth

Let $P_{t,i}$ and $Q_{t,i}$ denote the implicit price and real quantity for component i in quarter t . Real GDP growth is computed via the Fisher chain-weighted formula:

$$1 + g_t^{\text{GDP}} = \sqrt{\frac{\sum_i Q_{t,i} P_{t-1,i}}{\sum_i Q_{t-1,i} P_{t-1,i}} \cdot \frac{\sum_i Q_{t,i} P_{t,i}}{\sum_i Q_{t-1,i} P_{t,i}}}. \quad (22)$$

Imports enter the aggregation with a negative sign (GDP is domestic production).

10.1 Contributions to growth

GDPNow also reports contributions to GDP growth by component. Let $c_{t,i}$ denote the (non-annualized) contribution of component i . Then $g_t^{\text{GDP}} \approx \sum_i c_{t,i}$, and annualized contributions are reported in percentage points at SAAR.

11 Data requirements for replication

Replication requires three layers of inputs:

1. **Quarterly GDP component quantities and prices:** the 13 real components and their implicit deflators, for estimating quarterly BVARs and for Fisher aggregation.
2. **Monthly source data:** monthly indicators used by bridge equations and special modules, along with the price series needed to deflate them into “real” quantities.
3. **Large monthly panel:** the N -series dataset used for factor extraction (transformed to stationarity and standardized), with real-time jagged-edge handling.

12 Replication from the Atlanta Fed tracking workbook

The Atlanta Fed publishes an Excel workbook containing “underlying source data, forecasts, and model parameters” and forecast histories. Public GDPNow materials point users to tabs such as `TrackingHistory` (growth rates and changes) and `ContribHistory` (contributions) for complete histories.

If your goal is to reproduce the workbook’s archived GDP nowcast series exactly (e.g., the `TrackingArchives` tab, column AB labeled “GDP Nowcast”), you can treat the workbook as the authoritative source of intermediate inputs and focus on deterministic recomputation.

12.1 Deterministic recomputation check

Typical workbook-replication steps are:

1. Read the `TrackingArchives` row for a given “as-of” date and target quarter.
2. Extract the model-implied component forecasts used for aggregation (either component growth rates and level changes, or contributions).
3. If contributions are available, sum them to obtain the GDP nowcast; otherwise recompute GDP growth using Fisher chain-weighting from forecasted $P_{t,i}, Q_{t,i}$.
4. Compare your computed GDP nowcast to the workbook’s `GDP Nowcast` value (column AB).

13 Implementation outline in Python

There are two practical implementation targets:

1. **Spreadsheet replication (best for exact matching):** read the published tracking workbook, recompute the GDP nowcast mechanically, and verify it matches the archived nowcast.
2. **Full re-estimation (research replica):** build data pipelines from raw public releases, estimate the factor model and BVARs, run bridge equations and special modules, and compute Fisher aggregation.

13.1 Spreadsheet replication: recommended workflow

Algorithm 1 Workbook-based reproduction and validation

- 1: Load workbook with `pandas.read_excel(..., sheet_name=None)`
 - 2: Identify the `TrackingArchives` row corresponding to the chosen as-of date
 - 3: Extract the archived GDP nowcast (column AB) for that row
 - 4: Extract component forecasts needed for aggregation (growth rates and/or contributions)
 - 5: **if** component contributions are available **then**
 - 6: $\widehat{\text{GDPNowcast}} \leftarrow \sum_i \widehat{\text{contrib}}_i$
 - 7: **else**
 - 8: Build forecasted $\{P_{t,i}, Q_{t,i}\}$ and compute Fisher growth
 - 9: **end if**
 - 10: Compare computed nowcast to archived nowcast; report deviations
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Useful libraries. For workbook replication: `pandas`, `openpyxl`, `numpy`. For full re-estimation: add `statsmodels` (VAR), and a state-space/Kalman toolkit or implement a custom Kalman filter (the factor model is simple enough to code directly).

14 Selected documented model changes (2018–2025)

The public “Modifications to GDPNow” memo documents several important updates, including:

1. 2018: allow AR(1) idiosyncratic measurement errors for some factor-panel series.
2. 2020: remove outlier replacement in the factor panel.
3. 2022: add post-March-2020 dummies in FAAR monthly forecasting equations; adjust some bridge regressions and AR choices.
4. 2023: implement recency-weighted restricted LS when combining bridge and quarterly-BVAR forecasts; extend pandemic-handling adjustments to net exports and inventories.
5. 2025: adjust goods trade for gold; update deflators after BEA revisions (e.g., office structures data centers); impute missing price-index observations during the 2025 shutdown.

References

- [1] Patrick Higgins. *GDPNow: A Model for GDP “Nowcasting”*. Federal Reserve Bank of Atlanta Working Paper 2014–7, 2014.
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- [3] Federal Reserve Bank of Atlanta. *Pulling Back the Curtain on GDPNow*. Public explainer article, 2022.